

EXP-4	Design and train a model for objects detection with real time example
-------	---

AIM:

- To implement an object detection model using a pre-trained YOLOv5 network.
- To detect and label objects in a real-world image using bounding boxes.

Algorithm:

1. **Import Libraries:** Import required libraries such as PyTorch and OpenCV.
2. **Load YOLOv5 Model:** Use `torch.hub.load()` to download and load the YOLOv5 medium model (`yolov5m`).
3. **Read Input Image:** Load a real-world image using `cv2.imread()`.
4. **Inference:** Pass the image to the model to detect objects.
5. **Render Bounding Boxes:** Use `.render()` to draw bounding boxes and labels on the image.
6. **Visualize Output:** Concatenate the original and detected image and display using `cv2_imshow`.

Code:

```
import cv2
import torch
from google.colab.patches import cv2_imshow # Import cv2_imshow

# Load a pre-trained YOLOv5 model
model = torch.hub.load('ultralytics/yolov5', 'yolov5m')

# Load the image
image_path = '/content/stock-photo.jpg'

image = cv2.imread(image_path)

original = image.copy() # Assign a copy of the original image to original
```

```
# Perform inference on the image
results = model(image)

# Render results (bounding boxes and labels)
image_with_boxes = results.render()[0]

# Ensure both images have the same data type
original = original.astype(image_with_boxes.dtype)

side_by_side = cv2.hconcat([original, image_with_boxes])

# Display the side-by-side images using cv2_imshow
cv2_imshow(side_by_side)
```

Output:



Result:

The YOLOv5 object detection model successfully identified and labeled multiple objects within the input image. The output image displayed the bounding boxes and class labels over the detected objects.

